breaking a signal into sinusoids.

Deconvolution: The inverse operation of convolution: if $x[n] * h[n] = y[n]$, find $x[n]$ given only $h[n]$ and $y[n]$. Deconvolution is usually carried out by dividing the frequency spectra.

Delta encoding: A broad term referring to techniques that store data as the difference between adjacent samples. Used in ADC, data compression and many other applications.

Delta function: A normalized impulse. The discrete delta function is a signal composed of all zeros, except the sample at zero that has a value of one. The continuous delta function is similar, but more abstract.

Delta-sigma: Analog-to-digital conversion method popular in voice and music processing. Uses a very high sampling rate with only a single bit per sample, followed by decimation.

Dependent variable: In a signal, the dependent variable depends on the value of the indepen-dent variable. Example: when a voltage changes over time, time is the independent variable and voltage is the dependent variable.

Difference equation: Equation relating the past and present samples of the output signal with past and present samples of the input signal. Also called a *recursion equation*.

Dilation: A morphological operation. When applied to binary images, dilation makes the objects larger and can combine disconnected objects into a single object.

Discrete cosine transform (DCT): A relative of the Fourier transform. Decomposes a signal into cosine waves. Used in data compression.

Discrete derivative: An operation for discrete signals that is analogous to the derivative for continuous signals. A better name is the *first difference*.

Discrete Fourier transform (DFT): Member of the Fourier transform family dealing with time domain signals that are *discrete* and *periodic*.

Discrete integral: Operation on discrete signals that is analogous to the integral for continuous signals. A better name is the *running sum*.

Discrete signal: A signal that uses quantized variables, such as a digitized signal residing in a computer.

Discrete time Fourier transform (DTFT): Member of the Fourier transform family dealing with time domain signals that are *discrete* and *aperiodic*

Dithering: Adding noise to an analog signal before analog-to-digital conversion to prevent the digitized signal from becoming "stuck" on one value.

Domain: The independent variable of a signal. For example, a voltage that varies with time is in the *time domain*. Other common domains are the *spatial domain* (such as images) and the *frequency domain* (the output of the Fourier transform).

Double precision: A standard for floating point notation that used 64 bits to represent each number. See *single precision* for comparison.

DSP microprocessor: A type of microprocessor designed for rapid math calculations. Often has a pipeline and/or Harvard architecture. Also called a RISC.

Dynamic range: The largest amplitude a system can deal with divided by the inherent noise of the system. Also used to indicate the number of bits used in an ADC. Can also be used with parameters other than amplitude; see *frequency dynamic range*.

Edge enhancement: Any image processing algorithm that makes the edges more obvious. Also called a *sharpening* operation.

Edge response: In image processing, the output of a system when the input is an edge. The sharpness of the edge response is often used as a measure of the resolution of the system.

Elliptic filter: Used to separate one band of frequencies from another. Achieves a fast roll-off by allowing ripple in the passband and the stopband. Can be used in both analog and digital designs.

End effects: The poorly behaved ends of a filtered signal resulting from the filter kernel not being completely immersed in the input signal.

Erosion: A morphological operation. When applied to binary images, erosion makes the objects smaller and can break objects into two or more pieces.

Euler's relation: The most important equation in complex math, relating sine and cosine waves with